



MindVista HRMS

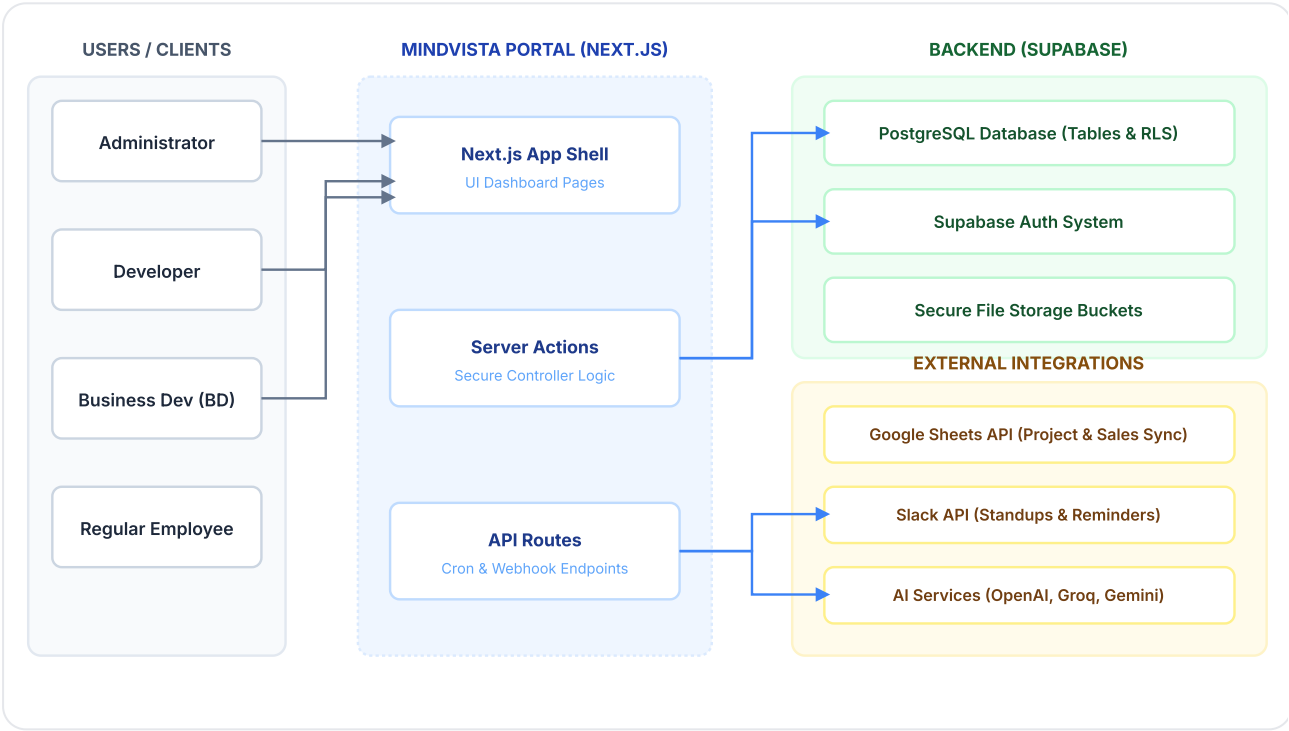
System Features & Implementation Documentation

1. System Overview

MindVista HRMS is an all-in-one Human Resource Management System built specifically for internal operations. The platform integrates traditional HR workflows with project management, a sales CRM, LinkedIn outreach tracking, automated payroll, and AI standup scoring.

- **Target Users:** Administrators, Team Leads, Business Developers, and Engineering Staff.
- **Core Goal:** Minimize spreadsheets and combine multiple business modules under a single portal.

System Architecture & Data Flow Diagram



2. User Roles and Permissions

The system uses a combination of a Primary Role and a PM (Project Management) Role to manage permissions.

Primary Roles

- **Admin (admin)**: Complete system access. Oversees employee creation, salary configurations, payroll calculations, and system settings.
- **Developer (developer)**: Extended administrative console access for debugging and management support.
- **Employee (employee)**: Standard user access to view profile, apply for leaves, check in/out, log timesheets, and view assigned projects/assets.

Project Management Roles (PM Role)

- **Business Development (bd)**: Identifies sales representatives. Used to grant access to the Sales CRM, LinkedIn outreach metrics, and client deal dashboards.
- **Developer (developer)**: Default classification for engineering and resource staff. Shows standard employee views.

Note: Project creation, editing, and deletion are managed directly through the primary Admin role.

Permission Matrix

Feature Module	Admin	Developer	BD (pm_role)	Employee
Employee Setup	Full Control	Full Control	No	No
Leave Approval	Yes	No	No	No
Attendance Logs	Full Access	Full Access	No	Self Only
Payroll Processing	Full + Approve	View Only	No	No
Project Creation	Yes	No	No	No
Sales CRM View	Yes	No	Yes	No
Asset Assignment	Yes	No	No	View Assigned

Feature Module	Admin	Developer	BD (pm_role)	Employee
Policies Upload	Yes	Read Only	Read Only	Read Only

3. Authentication and Security

- **Rate Limiting:** Protects `/login` by limiting attempts to 50 per email address in a 15-minute window.
- **JWT Verification:** Authenticates tokens locally before making database requests to prevent tampered token execution.
- **Audit Trails:** Immutable logs recorded in `auth_audit_log` track all sign-ins, logouts, and password resets with IP and user agent.
- **Pre-Onboarding Validation:** Multi-layer email checks (typo assistance, EmailVerify.io and MailboxValidator verification, DNS MX checks) and Pakistani CNIC format checks.

4. Dashboard

- **Employee View:** Displays leave balances, checking status, assigned assets, and a list of active personal projects.
- **Leads View:** Highlights a pending leave approval card containing requests from subordinates.
- **BD View:** Displays shortcuts to sales tracking (Daily Log, My Progress, ICP Filters) and outreach metrics.
- **Admin View:** Shows aggregate metrics (Total Projects, connection acceptance rate, and average engineering completion percentages).

5. Employee Management

- **Profile Registry:** Manages profile fields including name, email, CNIC number, date of birth, manager, lead, and work timings.
- **Employee Codes:** Unique 5-digit numbers auto-generated by database triggers upon user creation.
- **Permissions Enforces:** Core salary parameters (Basic Salary, Allowances, bank account info) are restricted to Admin updates only.
- **Self Service:** Employees can edit their phone number, home address, and emergency contact information from the profile tab.

6. Leave Management

- **Leave Types:** Sick Leave, Casual Leave, Annual Leave, and Unpaid Leave.
- **Deduction Engine:** Automatically excludes weekends (Saturdays, Sundays) and official calendar holidays from leave deductions.
- **Application Workflow:** Employee applies → System checks for overlaps and quota limits → Lead receives pending request.
- **Approval Actions:** Leads or Admins can approve (one-click) or reject (requires a mandatory explanation).

7. Attendance and Timesheets

- **Presence Tracking:** Daily check-in/out logging (Office, Remote, Half Day) with auto-computed decimal hours.
- **Timesheets:** Task-level logs for engineering metrics. Description text containing keywords like "bug", "fix", or "issue" are auto-counted as bug fixes.

8. Payroll and Compensation

- **Compensation History:** Versioned salary records. Creating a new version automatically dates and closes the previous one.
- **Commissions:** Configurable rules based on paid or invoiced values. Auto-calculated during the payroll cycle.
- **Calculation Engine:** Processes gross salaries, allowance packages, commissions, and unpaid leave deductions.
- **Anomaly Detection:** Flags warning and critical issues (e.g. missing compensation details) before final processing.
- **Approval & Delivery:** Approved cycles generate PDF slips and queue secure emails. Admin approves the queue to email slips to employees.
- **AI Assistant:** Controlled read-only assistant utilizing real payroll data to answer admin queries.

9. Performance Management

- **Goal Tracking:** Visual progress bars for goals. Progress can be updated manually or calculated automatically from timesheet and bug fix data.

- **Performance Reviews:** Formal quarterly review records (strengths, weaknesses, ratings) submitted by Admins.

10. Project Management

- **Project Records:** Stores details (value, status, priority, retainer status, timelines) synced from Google Sheets or imported via Excel templates.
- **Resource Allocation:** Assigns employees to projects with custom roles and allocation percentages.
- **Access Rules:** Admins see all projects. BDs see deals assigned to them. Engineers see projects where they are assigned as resources.

11. Sales and Business Development

- **Outreach Logs:** Daily tracking of connections sent, accepted invites, messages, and meetings booked per sales profile.
- **Sales Profiles:** Manages active platform profiles (LinkedIn, Upwork) and links them to spreadsheet IDs.
- **Command Center:** Admin dashboard showing live stats, individual rep targets, sheet snapshots, and weekly report summaries.

12. LinkedIn Intelligence and Outreach

- **Intelligence Import:** Parses user ZIP exports (connections, messages, skills) to calculate years of experience and top skills.
- **Outreach Dashboard:** Switch profiles, compare two profiles side by side, and view monthly, quarterly, or yearly campaign performance charts.
- **Monthly Report:** Headless browser automatically screenshots profile metrics at month-end, generates a PDF, and emails it to the Admin.
- **Slack Reminders:** Sends automated Slack notifications to reps to trigger data exports and outreach follow-ups.

13. ICP Filters (Ideal Customer Profile)

- **Target Criteria:** Tracks targeted job titles, companies, industry verticals, and regions.
- **Google Sheets Sync:** Pulls filter details directly from linked sheets.

- **Search & Match:** Full-text search engine with automatic geography extraction and duplicate detection.

14. Asset Management

- **Registry:** Tracks company equipment (laptops, monitors, mobile phones, and software licenses).
- **Lifecycle:** Asset registered (Available) → Assigned to employee (Assigned) → Employee returns equipment (Returned, status set back to Available) → History retained.

15. Company Policies

- **Library:** Document center for company handbooks, SOPs, NDA templates, and leave policies.
- **Storage:** Files are saved to private buckets in Supabase Storage with sanitized filenames. Admins upload and delete files; employees view or download them.

16. Holiday Calendar

- **Holiday Calendar:** Allows admins to define official company holidays.
- **Leave Adjustments:** Date checks automatically cross-reference this calendar to ensure leave balance days are not deducted during public holidays.

17. Notifications

- **In-App Alerts:** A bell icon in the top header displays unread notification counts.
- **Leave Requests:** Sends immediate alerts to employees when their leave requests are reviewed.

18. Daily Standups (Slack Integration)

- **Collection:** Imports standup posts from the Slack channel using API scopes.
- **AI Parser:** Extracts Completed Tasks, In Progress, and Blockers, and calculates performance scores.
- **Leaderboard & Trends:** Ranks employees by standup quality and plots average scores over a 6-week trend chart.

19. Team Hierarchy and Organization

- **Reporting Relationships:** Tracks managers and leads. The Lead handles leave approvals, while the Manager supervises performance.
- **Org Chart:** Generates a recursive visual org tree showing reporting hierarchies from any selected node.

20. Google Sheets Integration

- **Service Account:** Connects securely using credentials in environment variables.
- **Sheet Syncing:** Bi-directional sync of projects (incorporating name matching for BDs and managers), sales progress tracking, and ICP filters.

21. Automated Cron Jobs

Background jobs running on Vercel, secured via `CRON_SECRET` headers:

- `sync-standups` : Syncs standups from Slack and parses updates.
- `projects-sheet` : Daily import from the master Google Sheet.
- `linkedin-reminders` : Fires monthly, weekly, and follow-up alerts to the BD team.
- `monthly-report` : Generates and emails the LinkedIn analytics PDF.
- `payroll-reminder` : Notifies admins before pay day.
- `icp-filters` : Syncs ICP targeting criteria.

22. Email System

- **Transactional Emails:** Sends password reset links, verifications, and payroll alerts.
- **Salary Slips:** Slips are queued, approved by the admin, and sent with PDF attachments. Idempotency keys prevent double sending.

23. Technology Stack

- **Frontend:** Next.js 16 (App Router), React 19, Tailwind CSS v4, shadcn/ui, Recharts.
- **Backend:** Supabase (PostgreSQL database, Auth, Storage, RLS).
- **Integrations:** Google Sheets API, Slack API, OpenAI, Groq SDK, Puppeteer.

24. Deployment and Infrastructure

- **Hosting:** Vercel for the Next.js app and serverless cron schedulers.
- **Database & Storage:** Supabase handles data tables, auth sessions, and document storage buckets (`employee-documents` , `policies` , `assets-media`).
- **Environment Configuration:** Managed via secure backend keys for external services.